

CSA1013 – SOFTWARE ENGINEERING

Agile Sprint Planning & User Story Workshop

AI-Powered Smart Forest Fire Detection and Environmental Monitoring System

A Product Backlog, Sprint Planning, and User Story Workshop Report

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0. Introduction and Purpose of This Workshop

This document records the outcomes of an Agile Sprint Planning and User Story Workshop conducted for the AI-Powered Smart Forest Fire Detection and Environmental Monitoring System described in the accompanying Software Engineering project report. Where the original report establishes the problem statement, architecture, and justification for an Agile SDLC, this workshop translates that plan into the concrete Agile artefacts a Scrum team would actually work from: a prioritised product backlog, well-formed user stories with acceptance criteria, story-point estimates, and a committed sprint backlog with a clear sprint goal.

The workshop follows the standard Scrum cadence — backlog refinement, sprint planning, and the definitions that govern "ready" and "done" work — and is structured so that a development team could begin Sprint 1 immediately after this document is approved by the Product Owner. Two sprints are planned in detail (Sprint 1 and Sprint 2) so that the backlog's sequencing logic, dependency chain, and incremental value delivery are visible beyond a single iteration, and a lightweight release roadmap extends this view across the first program increment.

0.1 Agile Ceremonies Referenced in This Document

Ceremony	How It Is Reflected in This Document
Backlog Refinement	Section 4 (prioritisation) and Section 6 (estimation) capture the outputs of refinement sessions where the team and Product Owner size and order the backlog.
Sprint Planning	Section 5 and Section 8 define the Sprint 1 and Sprint 2 backlogs, the tasks each story is broken into, and the team's capacity commitment.
Daily Scrum	Not directly documented here (it is a daily, verbal ceremony), but the task breakdowns in Sections 5 and 8 are sized to be trackable at daily-standup granularity.
Sprint Review	Section 7 and Section 9 define the expected deliverables each sprint reviews against, and the stakeholders who attend.
Sprint Retrospective	Section 10 documents the retrospective format the team will use to inspect and adapt its process after each sprint.

0.2 Team Composition and Working Agreement

The Scrum team assumed for this planning exercise is a small, cross-functional group capable of covering the full sensing-to-alert data path: two backend engineers, one

frontend engineer, one machine-learning engineer, and one QA engineer, supported by a Scrum Master and a Product Owner proxy drawn from the Forest Fire Management Authority stakeholder group. This composition directly informs the capacity assumptions used in Sections 5 and 8.

Working Agreement Item	Team Commitment
Sprint length	Two weeks, with Sprint Planning on Day 1 and Sprint Review / Retrospective on Day 10.
Daily Scrum	15 minutes, same time each day; blockers raised immediately rather than held for standup.
Story ownership	Each story has a single accountable owner, even when multiple people contribute tasks.
Definition of Ready / Done	No story enters a sprint without meeting Section 6a; no story is marked complete without meeting Section 7.3 / 9's Definition of Done.
External dependencies	Any story depending on an external data provider (meteorological, satellite) is flagged at planning and, where possible, front-loaded early in the sprint.

0.3 Assumptions and Constraints

The following assumptions and constraints shape the backlog sequencing, estimates, and sprint commitments in this document, and should be revisited if project conditions change materially.

- The pilot deployment targets a single forest range, consistent with the phased rollout described in the project report's implementation roadmap.
- IoT hardware procurement and installation are treated as a parallel workstream; software stories assume sensor connectivity is available for integration testing by the relevant sprint.
- A baseline, pre-trained computer-vision model is assumed available for Sprint 1 integration (US-3.1); building a model from scratch would materially change that story's estimate.
- Story points are relative within this backlog only and are not intended to be compared against point scales from other teams or projects.
- Velocity of 22–26 points per sprint is an initial planning assumption; it will be recalibrated after Sprint 1's actual output is known.

1. Project Vision and Stakeholders

1.1 Project Vision

To transform forest fire management from a reactive, after-the-fact activity into a proactive, continuously monitored discipline by fusing IoT sensing, drone and satellite imagery, and AI-based analytics into a single early-warning and decision-support platform — enabling forest departments and emergency responders to detect ignitions while they are still small, and to act before they become uncontrollable.

1.2 Project Objectives

- Continuously monitor environmental conditions (temperature, humidity, smoke, wind) across forest zones using distributed IoT sensors.
- Detect fire signatures at an early, containable stage using AI-based computer vision and sensor fusion.
- Predict zone-level fire risk from historical, vegetation, and weather data.
- Extend coverage to remote and inaccessible terrain through drone and satellite integration.
- Deliver prioritised, low-latency alerts to forest officials, fire departments, and communities.
- Support firefighting operations with spread estimates and resource-deployment guidance.
- Provide reporting and analytics for long-term forest management and policy planning.

1.2b SMART Framing of the Vision

For backlog planning, the vision above is decomposed into SMART (Specific, Measurable, Achievable, Relevant, Time-bound) success conditions the team can trace individual epics against:

- Specific: detect a fire signature from ground sensor, drone, or satellite data rather than relying on visual confirmation alone.
- Measurable: generate a detection alert within 60 seconds of a confirmed anomaly, and keep dashboard queries under 2 seconds.
- Achievable: deliver the platform incrementally, pilot-first at a single forest range, before multi-zone scale-out.
- Relevant: every epic maps directly to a stakeholder responsibility identified in Section 1.3.

- Time-bound: a working detect-verify-alert loop is targeted for the end of Sprint 2 (Section 9), ahead of the pilot go-live milestone in the wider project roadmap.

1.3 Key Stakeholders

The stakeholder groups below drive backlog priorities and participate as Product Owner proxies, sprint reviewers, or end users of the delivered increments.

Stakeholder	Agile Role	Interest / Involvement in the Backlog
Forest Department / Government Environmental Agencies	Product Owner (policy)	Sets priority on compliance, coverage, and reporting features; approves release scope.
Forest Fire Management Authorities	Key Business Stakeholder	Validates AI-generated alerts; drives acceptance criteria for detection accuracy.
Fire & Emergency Response Services	End User	Consumes alerts, location data, and spread estimates during sprint reviews.
Forest Rangers & Field Officers	End User	Primary users of mobile alerts and field-verification stories; usability feedback.
Local & Indigenous Communities	Secondary End User	Receives early warnings; source of community-reporting user stories.
Meteorological & Climate Agencies	External Data Provider	Supplies weather/wind/drought-index feeds consumed by risk-prediction stories.
Environmental Research Organisations	Data Consumer	Uses historical data exports; informs reporting-module backlog items.
System Administrators	Technical Stakeholder / Scrum support	Owens non-functional stories: security, uptime, access control, deployment.

1.4 RACI Matrix for Backlog Governance

The matrix below clarifies decision rights over the backlog itself — who is Responsible for delivery, Accountable for priority calls, Consulted during refinement, and Informed at sprint review.

Activity	Forest Dept. Official	Fire Mgmt. Authority	Scrum Team	Field Ranger	System Admin
Set product vision & release scope	A	C	R	I	I
Prioritise product backlog	A	R	C	I	I
Write / refine user stories	C	C	R	C	C
Estimate story points	I	I	R/A	I	C
Approve sprint goal	A	C	R	I	I
Verify acceptance criteria	I	R	C	C	I
Own non-functional / security requirements	I	I	C	I	R/A

R = Responsible, A = Accountable, C = Consulted, I = Informed.

2. Epics and User Stories

The eight functional modules identified in the system architecture are treated as Epics. Each Epic is decomposed into user stories in the standard format: “As a <user>, I want <feature>, so that <benefit>.”

2.0 Epic Overview

Epic	Name	Primary Goal	# Stories
E1	User Management & Access Control	Secure, role-appropriate access to the platform.	5
E2	Environmental Monitoring	Continuous ground-truth visibility of forest conditions.	4
E3	Fire Detection (AI / Computer Vision)	Early, low-latency identification of fire signatures.	4
E4	Fire Risk Prediction	Forward-looking, explainable zone risk scoring.	4
E5	Drone & Satellite Integration	Extend coverage to remote and inaccessible terrain.	4

Epic	Name	Primary Goal	# Stories
E6	Reporting	Turn operational data into management and research insight.	3
E7	Notification & Alerting	Get the right alert to the right person, fast.	4
E8	Command Dashboard	One coherent operating picture for responders.	4

Epic 1 — User Management & Access Control

This Epic underpins every other Epic in the backlog: no environmental data, alert, or report should be visible to someone who isn't authorised to see it. It is therefore sequenced first in the product backlog (Section 4) and delivered in Sprint 1 (Section 5).

- *As a System Administrator, I want to create role-based user accounts, so that only authorised personnel can access sensitive location data or issue alerts.*
- *As a Forest Department Official, I want to log in securely with my credentials, so that I can access the dashboard appropriate to my role.*
- *As a System Administrator, I want to deactivate or reassign user roles, so that access stays current when personnel change.*
- *As a System Administrator, I want all login attempts to be audit-logged, so that I can investigate suspicious access.*
- *As a Forest Department Official, I want to reset my password through a verified channel, so that I regain access without administrator intervention for routine cases.*

Epic 2 — Environmental Monitoring

This Epic replaces the report's identified gap in continuous, ground-level sensing. It converts raw IoT telemetry into a live operating picture and is the primary data source both the detection Epic (3) and the risk-prediction Epic (4) build on.

- *As a Field Ranger, I want to view real-time temperature, humidity, smoke, and wind readings for my zone, so that I can assess ground conditions without visiting every site.*
- *As a System Administrator, I want failing or anomalous sensors flagged automatically, so that maintenance can be scheduled before data quality degrades.*
- *As a Forest Department Official, I want historical environmental trends for a zone, so that I can assess seasonal risk patterns.*

- *As a Field Ranger, I want to receive an in-app warning when readings in my zone cross a dangerous threshold (e.g., extreme dryness), so that I can act before an official fire alert is raised.*

Epic 3 — Fire Detection (AI / Computer Vision)

This is the platform's core value proposition and the highest-risk Epic in the backlog: it is the direct answer to the problem statement's central complaint — that fires are detected only after they have already spread. It carries the highest story-point estimate in Section 6 (US-3.1, 13 points) reflecting that risk.

- *As an Emergency Response Coordinator, I want the system to automatically flag early smoke or flame signatures from sensor and imagery data, so that I can respond before a fire spreads.*
- *As a Forest Fire Management Authority, I want to verify or dismiss an AI-generated fire alert with one action, so that false positives are quickly filtered out.*
- *As a System Administrator, I want detection models to be retrained on confirmed-incident feedback, so that accuracy improves each fire season.*
- *As an Emergency Response Coordinator, I want each detection alert to show a confidence score and the contributing sensor/imagery evidence, so that I can judge how urgently to act.*

Epic 4 — Fire Risk Prediction

Where Epic 3 answers "is a fire happening now," this Epic answers "where is a fire likely next," letting forest authorities pre-position patrols and drone coverage ahead of ignition rather than only reacting to it.

- *As a Forest Department Official, I want a zone-level fire-risk score updated regularly, so that I can plan patrols and pre-position resources.*
- *As a Meteorological Data Provider, I want my weather and drought-index feeds to be ingested automatically, so that risk scores reflect current conditions.*
- *As a Forest Fire Management Authority, I want to see the contributing factors behind a risk score, so that I can explain and trust the prediction.*
- *As a Forest Department Official, I want to compare risk scores across zones on a single ranked list, so that I can prioritise where to send patrols first.*

Epic 5 — Drone & Satellite Integration

Ground sensors alone cannot cover dense or mountainous terrain. This Epic extends the sensing layer vertically, fusing aerial and orbital data sources into the same detection and risk pipeline established by Epics 2–4.

- *As a Forest Fire Management Authority, I want drone patrols automatically scheduled over high-risk zones, so that remote areas receive extra coverage without manual planning.*
- *As a System Administrator, I want satellite thermal-anomaly feeds ingested into the detection pipeline, so that coverage extends beyond ground-sensor range.*
- *As an Emergency Response Coordinator, I want drone and satellite imagery fused with ground-sensor alerts on one map, so that I see a single situational picture.*
- *As a Forest Fire Management Authority, I want to manually request an on-demand drone flight over a specific coordinate, so that I can investigate an unconfirmed report immediately.*

Epic 6 — Reporting

Reporting turns the operational data generated by Epics 2–5 into the management and research insight identified in the stakeholder analysis, supporting long-term forest-management and policy decisions rather than only real-time response.

- *As a Forest Department Official, I want daily, weekly, and seasonal environmental and incident reports, so that I can inform policy and management decisions.*
- *As an Environmental Research Organisation, I want to export historical sensor and incident data, so that I can conduct ecological research.*
- *As a Forest Fire Management Authority, I want a detection-accuracy report comparing verified alerts to false positives, so that I can track model performance over each season.*

Epic 7 — Notification & Alerting

Detection and risk scoring only create value once the right person acts on them. This Epic is the delivery mechanism that turns an internal alert into a real-world response, and is prioritised immediately after Epic 3 in the backlog.

- *As a Field Ranger, I want to receive fire alerts via SMS and mobile app, so that I am notified even with limited connectivity.*
- *As a System Administrator, I want alert thresholds to be configurable per zone, so that sensitivity can be tuned to local conditions.*
- *As a Local Community Member, I want to receive early-warning notifications for my area, so that I can evacuate or take precautions in time.*
- *As an Emergency Response Coordinator, I want escalation reminders if a critical alert is not acknowledged within 5 minutes, so that no urgent alert is silently missed.*

Epic 8 — Command Dashboard

The dashboard is the single pane of glass that unifies every other Epic — monitoring, detection, risk, drone/satellite, and notifications — into one coherent operating picture, and is intentionally scheduled last since it depends on all upstream data being available.

- *As an Emergency Response Coordinator, I want a live map of sensor status, fire-risk heatmaps, and active incidents, so that I can coordinate a response from one screen.*
- *As a Forest Department Official, I want role-appropriate dashboard views, so that I only see the information relevant to my responsibilities.*
- *As a Field Ranger, I want the dashboard to load key alerts within 2 seconds on a mobile device, so that I can act quickly in the field.*
- *As an Emergency Response Coordinator, I want to filter the dashboard by zone, risk level, or incident status, so that I can focus on the areas that matter most right now.*

3. Acceptance Criteria

Acceptance criteria are defined in Given–When–Then format for the highest-priority user stories that anchor the first sprint and the core detection/alerting value chain.

US-1.2 — Secure login by role

- Given a registered user with valid credentials, When they submit their username and password on the login screen, Then they are authenticated and redirected to the dashboard view configured for their role.
- Given three consecutive failed login attempts, When a fourth attempt is made within 15 minutes, Then the account is temporarily locked and an audit-log entry is created.

US-2.1 — Real-time environmental readings

- Given a zone with active IoT sensors, When a new reading is transmitted, Then the dashboard reflects the updated temperature, humidity, smoke, and wind values within 5 seconds.
- Given a sensor that stops transmitting for more than 10 minutes, When the timeout threshold is reached, Then the sensor is marked "Offline" and flagged for maintenance.

US-3.1 — Early smoke/flame detection alert

- Given fused sensor and imagery data for a zone, When the AI model confidence for a fire signature exceeds the configured threshold, Then a fire-detection alert is generated and logged within 60 seconds of the confirmed anomaly.
- Given an active fire-detection alert, When a Forest Fire Management Authority marks it "Verified" or "False Positive," Then the outcome is stored and used as feedback for the next model-retraining cycle.

US-4.1 — Zone-level fire-risk score

- Given current weather, vegetation-dryness, and historical fire data for a zone, When the scheduled risk-scoring job runs, Then an updated risk score (Low/Medium/High/Extreme) is published for that zone with a timestamp.
- Given a published risk score, When a user opens the zone detail view, Then the top three contributing factors to that score are displayed.

US-5.1 — Automatic drone patrol scheduling

- Given a zone whose risk score changes to High or Extreme, When the scheduling engine runs its next cycle, Then a drone patrol is automatically queued for that zone and appears on the flight schedule.

US-7.1 — Multi-channel fire alert notification

- Given a confirmed fire-detection alert for a zone, When the alert is generated, Then SMS, email, and in-app notifications are dispatched to all subscribed users for that zone within 60 seconds.
- Given a user with no configured notification channel, When an alert is generated for their zone, Then the system logs a delivery-configuration warning for the administrator.

US-8.1 — Live command dashboard

- Given an authenticated user with dashboard access, When they open the Command Dashboard, Then live sensor status, the fire-risk heatmap, and active incidents load within 2 seconds.
- Given a Field Ranger viewing the dashboard on a mobile device, When a new alert is issued for their assigned zone, Then it appears at the top of their alert feed within 2 seconds of generation.

US-2.2 — Automatic faulty-sensor flagging

- Given a sensor reporting values outside its calibrated physical range, When three consecutive out-of-range readings are received, Then the sensor is flagged "Anomalous" and a maintenance ticket is created.

US-4.2 — Weather / drought-index feed ingestion

- Given a scheduled sync with the meteorological data provider, When the sync completes successfully, Then the latest weather and drought-index values are available to the risk-scoring job within 15 minutes.
- Given the meteorological feed is unavailable, When the risk-scoring job runs, Then it uses the last known values and flags the score as "stale data" on the dashboard.

US-5.2 — Satellite thermal-feed ingestion

- Given a new satellite pass over a monitored zone, When thermal-anomaly data is published by the provider, Then it is ingested and geo-aligned with the zone boundary within one polling cycle.

US-6.1 — Scheduled analytical reports

- Given the end of a reporting period (day/week/season), When the scheduled report job runs, Then a report covering environmental trends, incidents, and detection accuracy is generated and made available for download.

US-7.2 — Configurable alert thresholds

- Given an administrator editing a zone's alert configuration, When they save a new threshold value, Then subsequent alert evaluations for that zone use the updated threshold immediately.

US-8.2 — Role-appropriate dashboard views

- Given a user with the Field Ranger role, When they open the dashboard, Then only their assigned zones and field-relevant widgets are shown, excluding administrative controls.

US-1.3 — Password reset via verified channel

- Given a user who has forgotten their password, When they request a reset via their registered email or phone, Then a time-limited reset link/code is issued and the old password is invalidated once a new one is set.

US-2.3 — In-app threshold warning for rangers

- Given live readings for a ranger's assigned zone, When a reading crosses the configured danger threshold, Then the ranger receives an in-app push warning within 30 seconds, distinct in styling from a confirmed fire alert.

US-3.3 — Confidence score and evidence on alerts

- Given a generated fire-detection alert, When a user opens its detail view, Then the model's confidence score and the specific sensor readings or imagery frames that triggered it are displayed.

US-4.3 — Cross-zone risk ranking view

- Given risk scores for multiple zones, When a user opens the ranking view, Then zones are listed in descending order of current risk score, refreshed on the same cadence as the underlying scores.

US-5.3 — On-demand drone flight request

- Given a Forest Fire Management Authority user with drone-request permission, When they submit coordinates for an on-demand flight, Then the request is queued and the requester is notified once imagery is available.

US-6.2 — Detection-accuracy report

- Given a completed reporting period, When the detection-accuracy report is generated, Then it shows the ratio of verified to false-positive alerts and trends this ratio against the previous period.

US-7.3 — Unacknowledged-alert escalation

- Given a critical alert with no acknowledgement, When 5 minutes elapse, Then the alert is automatically escalated to the next responder tier and the escalation is logged.

US-8.3 — Dashboard filtering by zone, risk, or status

- Given a user viewing the Command Dashboard, When they apply a zone, risk-level, or incident-status filter, Then the map, heatmap, and incident list update to reflect only the filtered results without a full page reload.

4. Product Backlog Prioritisation

The backlog is prioritised using MoSCoW categorisation combined with a relative business-value score (1–10, set by stakeholder proxies) and dependency analysis — e.g., detection alerts depend on environmental data ingestion and authentication, so foundational stories are ranked above the AI-heavy features that consume them.

ID	User Story (summary)	Epic	MoSCoW	Value (1–10)	Points	Backlog Rank
US-1.1	Role-based account creation	Epic 1	Must	8	3	1
US-1.2	Secure role-based login	Epic 1	Must	9	3	2
US-1.3	Password reset via verified channel	Epic 1	Should	5	3	16

ID	User Story (summary)	Epic	MoSCoW	Value (1–10)	Points	Backlog Rank
US-2.1	Real-time environmental readings view	Epic 2	Must	9	5	3
US-2.2	Automatic faulty-sensor flagging	Epic 2	Should	6	5	10
US-2.3	In-app threshold warning for rangers	Epic 2	Should	6	3	13
US-3.1	Early smoke/flame AI detection alert	Epic 3	Must	10	13	4
US-3.2	Alert verification / false-positive dismissal	Epic 3	Must	8	5	5
US-3.3	Confidence score & evidence on alerts	Epic 3	Should	7	5	9
US-4.1	Zone-level fire-risk scoring	Epic 4	Must	9	8	6
US-4.2	Weather/drought feed ingestion	Epic 4	Should	7	5	11
US-4.3	Cross-zone risk ranking view	Epic 4	Could	5	3	18
US-5.1	Automatic drone patrol scheduling	Epic 5	Should	6	8	14
US-5.2	Satellite thermal-feed ingestion	Epic 5	Should	6	8	15
US-5.3	On-demand drone flight request	Epic 5	Could	4	5	19
US-6.1	Scheduled analytical reports	Epic 6	Could	5	5	17
US-6.2	Detection-accuracy report	Epic 6	Could	4	5	20
US-7.1	Multi-channel emergency notification	Epic 7	Must	9	8	8

ID	User Story (summary)	Epic	MoSCoW	Value (1–10)	Points	Backlog Rank
US-7.2	Configurable alert thresholds	Epic 7	Should	6	3	12
US-7.3	Unacknowledged-alert escalation	Epic 7	Should	7	5	7
US-8.1	Live command dashboard (map, heatmap, incidents)	Epic 8	Must	9	13	21
US-8.2	Role-appropriate dashboard views	Epic 8	Should	6	5	22
US-8.3	Dashboard filtering by zone/risk/status	Epic 8	Could	5	3	23

Ranking logic: Epic 1 (identity) and Epic 2 (data ingestion) are sequenced first because every other Epic depends on authenticated access and live sensor data. Epic 3 (detection) and Epic 7 (notification) follow immediately, since they deliver the platform's core value proposition — early warning. Epic 4 (risk prediction) is scheduled alongside detection as it shares the same data pipeline. Epics 5, 6, and 8 extend coverage, insight, and usability once the core detect-and-alert loop is proven.

5. Sprint Backlog — Sprint 1

5.1 Sprint 1 Scope

Sprint 1 (2-week sprint) selects the top five backlog items that establish the authentication, data-ingestion, and core-detection foundation the rest of the platform depends on.

ID	User Story	Development Tasks
US-1.1	Role-based account creation	Design user/role schema; implement registration API; build admin UI for account creation; write unit tests.
US-1.2	Secure role-based login	Implement login API with hashed credentials; configure role-based route guards; build login screen; add audit-log entries.
US-2.1	Real-time environmental readings view	Define sensor-reading schema; build MQTT ingestion listener; implement

ID	User Story	Development Tasks
		time-series write path; build live-readings widget on dashboard.
US-3.1	Early smoke/flame AI detection alert	Integrate baseline smoke/flame detection model into ingestion pipeline; implement alert-generation service; log alert with confidence score; verify sub-60-second latency in test environment.
US-3.2	Alert verification / false-positive dismissal	Build alert-verification action (Verified / False Positive) in the operator UI; persist outcome; expose feedback dataset for future retraining.

5.2 Sprint Capacity Note

Total committed effort for Sprint 1 is 29 story points (3 + 3 + 5 + 13 + 5 – see Section 6). Given a small cross-functional team (2 backend, 1 frontend, 1 ML engineer, 1 QA) at an assumed velocity of 22–26 points per two-week sprint, this is a deliberately ambitious first commitment; US-3.1's 13 points reflect the added uncertainty of integrating an unproven detection model, and the team has agreed to timebox model-integration spikes so the sprint goal is not put at risk. Should US-3.1 run over, US-3.2 is the agreed candidate to descope into Sprint 2 without breaking the sprint goal.

5.3 Sprint 1 Burndown Plan (Ideal)

The table below is the ideal burndown the team tracks against at each daily scrum; deviations trigger a conversation about scope, blockers, or re-estimation rather than silent schedule slippage.

Day	Remaining Points (Ideal)	Key Milestone	Owner Focus
Day 1	29	Sprint planning complete; environment set up	Whole team
Day 3	23	Auth schema & API scaffolded	Backend
Day 5	18	Login UI + MQTT listener functional	Frontend / Backend

Day	Remaining Points (Ideal)	Key Milestone	Owner Focus
Day 7	12	Detection-model spike results reviewed	ML engineer
Day 9	6	Alert generation + verification UI integrated	Full stack
Day 10	0	Sprint review demo & retrospective	Whole team

6a. Definition of Ready

Before a story is pulled into a sprint, it must satisfy the following Definition of Ready — agreed by the team so that Sprint Planning is spent committing to work, not discovering unknowns.

- The story follows the "As a / I want / so that" format and names a specific user role.
- Acceptance criteria are written in Given–When–Then form and reviewed by the Product Owner proxy.
- The story has been estimated by the team using Planning Poker.
- Dependencies on other stories or external data providers (e.g., meteorological feeds) are identified and, if blocking, resolved or sequenced correctly.
- The story is small enough to be completed within a single sprint; oversized stories are split before entering the backlog.

6. Story Point Estimation

Story points are estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), reflecting relative effort, complexity, and uncertainty rather than absolute hours.

ID	User Story	Points	Estimation Justification
US-1.1	Role-based account creation	3	Well-understood CRUD + role model; low technical uncertainty; small UI surface.

ID	User Story	Points	Estimation Justification
US-1.2	Secure role-based login	3	Standard authentication pattern; slight added effort for lockout/audit logging.
US-1.3	Password reset via verified channel	3	Standard email/SMS token flow; low complexity, well-precedented pattern.
US-2.1	Real-time environmental readings view	5	Requires a new ingestion path (MQTT → time-series DB → live widget); moderate integration risk.
US-2.2	Automatic faulty-sensor flagging	5	Needs anomaly-detection logic and threshold tuning across heterogeneous sensor types.
US-2.3	In-app threshold warning for rangers	3	Reuses the readings pipeline; adds a rule check and a push notification.
US-3.1	Early smoke/flame AI detection alert	13	Highest uncertainty: model integration, latency target (60s), and sensor/imagery fusion; team has not built this pipeline before.
US-3.2	Alert verification / false-positive dismissal	5	New UI workflow plus a feedback-capture data model; moderate complexity.
US-3.3	Confidence score & evidence on alerts	5	Requires surfacing model internals and linked source data; moderate UI/backend coupling.
US-4.1	Zone-level fire-risk scoring	8	Multiple data sources (weather, vegetation, history) must be combined into a scheduled scoring job.
US-4.2	Weather/drought feed ingestion	5	External API integration with a well-documented provider; moderate mapping effort.
US-4.3	Cross-zone risk ranking view	3	Simple read-model over existing risk scores; mostly presentation work.

ID	User Story	Points	Estimation Justification
US-5.1	Automatic drone patrol scheduling	8	New scheduling engine plus dependency on live risk scores; cross-Epic coordination.
US-5.2	Satellite thermal-feed ingestion	8	External feed with unfamiliar format; parsing and geo-alignment add complexity.
US-5.3	On-demand drone flight request	5	New manual-trigger workflow layered on the existing scheduling engine.
US-6.1	Scheduled analytical reports	5	Templating and export logic over existing data; low technical risk.
US-6.2	Detection-accuracy report	5	Aggregates verification outcomes from US-3.2; moderate reporting-logic complexity.
US-7.1	Multi-channel emergency notification	8	Three delivery channels (SMS/email/app) with retry logic and delivery guarantees within 60s.
US-7.2	Configurable alert thresholds	3	Simple configuration UI and rule evaluation; low complexity.
US-7.3	Unacknowledged-alert escalation	5	Requires a timer/escalation-chain service and acknowledgement tracking.
US-8.1	Live command dashboard	13	Composite view aggregating map, heatmap, and incident feed; performance target of 2s adds risk.
US-8.2	Role-appropriate dashboard views	5	Depends on US-8.1 and the role model; moderate conditional-rendering effort.
US-8.3	Dashboard filtering by zone/risk/status	3	Client-side filtering over an existing data set; low complexity.

Total estimated backlog effort: 132 story points across 23 user stories. At an assumed team velocity of 22–26 points per two-week sprint, the estimated backlog above

represents roughly five to six sprints (10–12 weeks) of delivery — consistent with the Foundation, Core Feature, and Optimisation sprint phases in the wider project roadmap.

7. Sprint Goal and Expected Deliverables

7.1 Sprint 1 Goal

“By the end of Sprint 1, authorised users can securely log in by role, view live environmental readings for their assigned zone, and receive an AI-generated early fire-detection alert that a Forest Fire Management Authority can verify or dismiss — establishing the core sense-detect-verify loop the rest of the platform builds on.”

7.2 Expected Sprint Deliverables

- A working authentication service with role-based access control and audit logging (US-1.1, US-1.2).
- A live environmental-monitoring view showing temperature, humidity, smoke, and wind data streamed from IoT sensors (US-2.1).
- A first-version AI detection service that raises a fire-detection alert within 60 seconds of a confirmed anomaly (US-3.1).
- An operator workflow for verifying or dismissing detection alerts, with outcomes captured for future model retraining (US-3.2).
- Automated unit and integration tests covering the authentication and ingestion-to-alert data path.
- A sprint review demo to Forest Fire Management Authority representatives, and a refined backlog for Sprint 2 (risk prediction and notification).

7.3 Definition of Done

- Code merged, peer-reviewed, and passing CI (unit + integration tests).
- Acceptance criteria for each story verified against the Given–When–Then conditions in Section 3.
- Feature demonstrated in the sprint review and accepted by the Product Owner / stakeholder proxy.
- No critical or high-severity defects open against the sprint's stories.
- Documentation updated (API contracts, data schema, deployment notes).

8. Sprint Backlog — Sprint 2

Sprint 2 builds directly on Sprint 1's sense-detect-verify loop by adding risk prediction and reliable multi-channel alerting — the next-highest-ranked items in Section 4 whose dependencies (authentication, live sensor data, detection pipeline) are now satisfied.

8.1 Sprint 2 Scope

ID	User Story	Development Tasks
US-4.1	Zone-level fire-risk scoring	Implement scheduled risk-scoring job combining weather, vegetation, and historical data; expose risk-score API; add contributing-factor breakdown.
US-4.2	Weather/drought feed ingestion	Integrate meteorological provider API; map fields to internal schema; add stale-data fallback handling.
US-7.1	Multi-channel emergency notification	Build notification-dispatch service (SMS/email/in-app); implement per-channel retry logic; wire to detection-alert events.
US-7.3	Unacknowledged-alert escalation	Add acknowledgement tracking to alerts; implement escalation timer service; configure 5-minute escalation rule.
US-2.2	Automatic faulty-sensor flagging	Implement anomaly-detection rule for sensor readings; auto-flag sensors after three out-of-range readings; create maintenance-ticket integration.

8.2 Sprint 2 Capacity Note

Total committed effort for Sprint 2 is 29 story points (8 + 5 + 8 + 5 + 5 – see Section 6), matching Sprint 1's actual capacity assuming velocity holds. The meteorological-feed dependency (US-4.2) is flagged as an external-provider risk during planning; the team agrees to begin that integration first so any provider-side delay surfaces early rather than blocking the sprint goal near the end of the iteration.

9. Sprint 2 Goal and Expected Deliverables

9.1 Sprint 2 Goal

“By the end of Sprint 2, every zone has a live, explainable fire-risk score driven by current weather and historical data, and every confirmed fire alert reaches responders through SMS, email, and in-app channels — with automatic escalation if nobody acknowledges it within five minutes.”

9.2 Expected Sprint 2 Deliverables

- A scheduled fire-risk scoring engine publishing zone-level scores with contributing factors (US-4.1).
- Live ingestion of weather and drought-index data, with graceful degradation when the feed is unavailable (US-4.2).
- A multi-channel notification service (SMS, email, in-app) with delivery retries for confirmed fire alerts (US-7.1).
- An escalation mechanism that re-notifies or reassigns unacknowledged critical alerts (US-7.3).
- Automatic flagging of failing or anomalous sensors, reducing false readings feeding the risk and detection engines (US-2.2).
- A sprint review demonstrating the end-to-end chain: sensor reading → risk score / detection alert → multi-channel notification → escalation.

10. Sprint Retrospective Format

At the close of each sprint, the team holds a time-boxed (45–60 minute) retrospective using the Start / Stop / Continue format, facilitated by the Scrum Master. Outcomes are logged as backlog items (process improvements) or carried into the team's working agreement.

Prompt	Guiding Question
Start	What should the team begin doing that it isn't doing now (e.g., earlier spikes for AI-model stories, clearer NFR acceptance criteria)?
Stop	What is creating friction or waste (e.g., underestimating external-integration stories, insufficient test data for detection models)?
Continue	What worked well and should be preserved (e.g., daily sync with the Forest Fire Management Authority proxy, pairing on the detection pipeline)?

For this project specifically, the retrospective after Sprint 1 is expected to surface at least one process action around AI-model story estimation, since US-3.1 carries the

highest uncertainty in the backlog; the team should decide whether future model-integration stories need a dedicated "spike" story ahead of the implementation story.

11. Release Roadmap Across Sprints

The table below extends the two detailed sprints into an indicative release plan for the remaining backlog, aligning with the Foundation → Core Feature → Optimisation sprint phases described in the wider project report.

Sprint	Theme	Primary Backlog Items
Sprint 1	Foundation	US-1.1, US-1.2, US-2.1, US-3.1, US-3.2 — identity, live sensing, first detection loop.
Sprint 2	Foundation → Core	US-4.1, US-4.2, US-7.1, US-7.3, US-2.2 — risk scoring and reliable alerting.
Sprint 3	Core Feature	US-1.3, US-2.3, US-3.3, US-7.2, US-4.3 — usability and transparency refinements.
Sprint 4	Optimisation	US-5.1, US-5.2 — drone scheduling and satellite ingestion extend coverage.
Sprint 5	Optimisation	US-5.3, US-6.1, US-6.2 — on-demand drones and analytical reporting.
Sprint 6	Hardening	US-8.1, US-8.2, US-8.3 — the full command dashboard, security and performance hardening.

This roadmap is indicative and will be re-sequenced at each sprint review based on actual velocity, stakeholder feedback, and any re-prioritisation the Product Owner makes in response to pilot-deployment findings.

11a. Non-Functional Requirements as Backlog Items

Non-functional requirements from the source report (reliability, security, scalability, performance) are not delivered as a single story; they are woven into the acceptance criteria of the relevant functional stories and additionally tracked as their own lightweight backlog items so they are never silently dropped under sprint pressure.

NFR	Backlog Treatment	Primary Related Stories	Points
Reliability (99.5% uptime)	Cross-cutting acceptance criterion on all sensing/alerting	US-2.1, US-7.1	5

NFR	Backlog Treatment	Primary Related Stories	Points
	stories; monitored via uptime dashboard story.		
Security (encryption, RBAC)	Embedded in every Epic-1 story; audited as part of Sprint hardening.	US-1.1, US-1.2	8
Performance (60s alert, 2s dashboard)	Explicit Given–When–Then criteria on US-3.1 and US-8.1.	US-3.1, US-8.1	included above
Scalability (single range → multi-zone)	Architectural constraint validated in Sprint 6 hardening; not a standalone story.	US-8.1	5
Data retention (5 years)	Storage-layer backlog item scheduled alongside Reporting Epic.	US-6.1	3

11b. Sprint-Level Risk Register

In addition to the project-wide risk table in the source report, the following risks are tracked specifically against Sprint 1 and Sprint 2 delivery.

Risk	Description	Likelihood	Mitigation
Model integration overrun	US-3.1's detection-model integration takes longer than the 13-point estimate allows.	Medium	Time-boxed spike in the first three days; US-3.2 pre-agreed as descope candidate.
External feed instability	Meteorological or satellite provider APIs are unavailable or change format.	Medium	Stale-data fallback built into US-4.2 acceptance criteria; provider integration front-loaded in Sprint 2.
Sensor connectivity gaps	Pilot-site IoT hardware is not fully installed when Sprint 1 begins.	Low–Medium	Use simulated sensor-data feed for development and testing until

Risk	Description	Likelihood	Mitigation
			hardware is confirmed live.
Stakeholder availability	Forest Fire Management Authority proxy unavailable for sprint review / acceptance sign-off.	Low	Sprint review scheduled a week in advance; async video walkthrough as fallback.

11c. Story Map — User Activities to Releases

A story map orders backlog items along two axes: the user's end-to-end journey (the backbone, left to right) and release priority (top to bottom). It gives a Product Owner a single view of what a user can do at each release, independent of which Epic a story technically belongs to.

User Journey Step →	Log In	See Conditions	Get Warned	Verify & Respond	Review & Improve
Release 1 (Sprint 1–2)	US-1.1, US-1.2	US-2.1	US-3.1, US-4.1	US-3.2, US-7.1	—
Release 2 (Sprint 3)	US-1.3	US-2.3	US-3.3, US-4.2	US-7.2, US-4.3	—
Release 3 (Sprint 4–5)	—	US-5.2	US-5.1	US-5.3	US-6.1, US-6.2
Release 4 (Sprint 6)	—	US-8.2, US-8.3	US-8.1	US-7.3	—

Reading down any column shows how a single capability deepens release over release — for example, the "Get Warned" column moves from a first working detection alert and risk score (Release 1), to richer weather-aware risk data (Release 2), to drone-extended coverage (Release 3), to a fully consolidated dashboard experience (Release 4).

11d. Sample Sprint 1 Task Board (Illustrative)

The board below illustrates how Sprint 1's five stories (Section 5) decompose into a Kanban-style task board at the start of the sprint, for use in the daily scrum.

To Do	In Progress	In Review	Done
Build admin UI for account creation (US-1.1)	Design user/role schema (US-1.1)	—	—
Configure role-based route guards (US-1.2)	Implement login API (US-1.2)	—	—
Build live-readings widget (US-2.1)	Build MQTT ingestion listener (US-2.1)	Define sensor-reading schema (US-2.1)	—
Log alert with confidence score (US-3.1)	Model-integration spike (US-3.1)	—	—
Persist verification outcome (US-3.2)	—	—	—

This snapshot represents Day 2 of the sprint: the sensor-reading schema (US-2.1) has already moved to review, while the highest-risk item — the detection-model spike (US-3.1) — is deliberately pulled into progress early so any surprises surface with maximum runway left in the sprint.

11e. Appendix — Full User Story Index

For quick reference during backlog refinement and sprint planning, the table below indexes every user story in this document by ID, Epic, and one-line description.

ID	Epic	One-Line Description
US-1.1	E1	Role-based account creation by an administrator.
US-1.2	E1	Secure role-based login for all user types.
US-1.3	E1	Self-service password reset via a verified channel.
US-2.1	E2	Real-time view of temperature, humidity, smoke, and wind per zone.
US-2.2	E2	Automatic flagging of failing or anomalous sensors.

ID	Epic	One-Line Description
US-2.3	E2	In-app threshold warnings for field rangers ahead of a confirmed alert.
US-3.1	E3	Early AI-based smoke/flame detection alert generation.
US-3.2	E3	Human verification or dismissal of a detection alert.
US-3.3	E3	Confidence score and supporting evidence shown on each alert.
US-4.1	E4	Scheduled zone-level fire-risk scoring.
US-4.2	E4	Ingestion of external weather / drought-index data.
US-4.3	E4	Cross-zone risk ranking view for prioritising patrols.
US-5.1	E5	Automatic drone-patrol scheduling for high-risk zones.
US-5.2	E5	Satellite thermal-anomaly feed ingestion.
US-5.3	E5	Manual, on-demand drone flight request.
US-6.1	E6	Scheduled daily / weekly / seasonal analytical reports.
US-6.2	E6	Detection-accuracy report tracking verified vs. false-positive alerts.
US-7.1	E7	Multi-channel (SMS / email / in-app) emergency notification.
US-7.2	E7	Configurable, per-zone alert thresholds.
US-7.3	E7	Automatic escalation of unacknowledged critical alerts.
US-8.1	E8	Live command dashboard — map, heatmap, and incident tracking.

ID	Epic	One-Line Description
US-8.2	E8	Role-appropriate dashboard views.
US-8.3	E8	Dashboard filtering by zone, risk level, or incident status.

13. Summary and Next Steps

This workshop has produced a complete, traceable Agile planning chain for the AI-Powered Smart Forest Fire Detection and Environmental Monitoring System: a vision and stakeholder map (Section 1), eight Epics decomposed into 23 user stories (Section 2), Given–When–Then acceptance criteria for the highest-priority stories (Section 3), a prioritised and estimated product backlog (Sections 4 and 6), two fully planned sprints with committed sprint goals (Sections 5, 7, 8, and 9), and the supporting governance artefacts — Definition of Ready, Definition of Done, retrospective format, risk register, and release roadmap — a Scrum team needs to execute against it.

The immediate next step is Product Owner sign-off on the Sprint 1 goal and backlog (Section 5 and 7), followed by environment setup and the first Sprint Planning ceremony. Subsequent sprints should re-run the estimation and prioritisation exercises in Sections 4 and 6 as real velocity data becomes available, rather than treating this document's figures as fixed.

14. Glossary of Agile Terms Used in This Document

Term	Definition
Epic	A large body of work that can be broken down into a number of smaller user stories.
User Story	A short, plain-language description of a feature from the perspective of the person who wants it.
Acceptance Criteria	The conditions a user story must satisfy to be considered complete, often written in Given–When–Then form.
Product Backlog	The single, ordered list of everything known to be needed in the product.
Sprint Backlog	The subset of product backlog items selected for a sprint, plus a plan for delivering them.
Story Points	A relative, unit-less measure of the effort, complexity, and uncertainty of a user story.

Term	Definition
Velocity	The average number of story points a team completes per sprint, used to forecast future capacity.
Sprint Goal	A short statement of what the sprint is meant to achieve, giving the team a shared purpose.
Definition of Ready	The agreed checklist a story must meet before it can be pulled into a sprint.
Definition of Done	The agreed checklist a story must meet before it is considered complete.
MoSCoW	A prioritisation technique categorising items as Must have, Should have, Could have, or Won't have.
Burndown Chart	A chart showing remaining work in a sprint over time, used to track progress against the plan.
Planning Poker	A consensus-based estimation technique where team members privately select a story-point card, then reveal and discuss simultaneously.
Spike	A short, time-boxed investigation used to reduce uncertainty on a story before committing to its full estimate.
RACI Matrix	A responsibility-assignment chart identifying who is Responsible, Accountable, Consulted, and Informed for a given activity.
Story Map	A two-dimensional arrangement of user stories along a user's journey (backbone) and release priority (depth).
Kanban Board	A visual board (To Do / In Progress / Review / Done) used to track the flow of work items during a sprint.
Product Owner	The role accountable for maximising product value and ordering the product backlog.
Scrum Master	The role responsible for facilitating Scrum events and helping the team follow its own working agreement.